

BAYAN-OVOO, Mongolia

WMO: 443020

Lat: 47.7858N

Lon: 112.1106E

Elev: 3038

StdP: 13.15

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-23.2	-32.5	1.0	-26.7	-28.5	1.2	-22.6	27.6	2.0	24.8	6.4	6.1	270	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	90.0	63.0	85.9	61.7	82.1	60.7	66.8	81.7	65.1	78.7	63.5	76.2	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	94.7	71.0	60.7	88.7	69.3	59.0	83.3	68.7	33.2	82.0	31.8	78.6	30.6	76.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.7	23.3	20.8	-32.4	94.9	3.5	4.6	-35.0	98.2	-37.1	100.9	-39.0	103.5	-41.6	106.8		
			-32.8	70.0	3.6	2.1	-35.4	71.6	-37.5	72.8	-39.5	74.0	-42.1	75.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	-6.4	1.5	18.9	38.9	52.2	63.5	68.6	64.4	52.6	35.3	13.0	-2.0	
	DBStd	27.95	9.02	10.38	12.35	9.01	8.56	7.42	5.75	6.50	8.45	9.27	12.64	9.51	
	HDD50	7744	1748	1357	964	348	76	2	0	1	65	462	1110	1611	
	HDD65	11774	2213	1777	1429	783	407	113	25	89	380	922	1560	2076	
	CDD50	1740	0	0	0	15	144	407	577	448	144	5	0	0	
	CDD65	295	0	0	0	0	9	68	137	72	9	0	0	0	
	CDH74	3711	0	0	0	10	235	1007	1465	837	157	0	0	0	
	CDH80	1292	0	0	0	2	61	368	552	276	33	0	0	0	
Wind	WSAvg	8.5	9.1	8.5	8.7	10.6	10.6	8.1	6.6	6.4	8.0	8.2	8.1	9.1	
Precipitation	PrecAvg	11.1	0.1	0.1	0.1	0.4	0.6	1.8	3.1	2.9	1.2	0.4	0.2	0.1	
	PrecMax	17.6	0.2	0.3	0.7	1.5	1.3	4.4	6.9	6.6	3.8	1.1	0.9	0.4	
	PrecMin	7.4	0.0	0.0	0.0	0.0	0.0	0.0	1.1	0.7	0.1	0.0	0.0	0.0	
	PrecStd	2.2	0.1	0.1	0.2	0.4	0.4	1.1	1.4	1.4	0.9	0.3	0.2	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	35.7	57.9	74.1	86.5	94.2	96.7	92.0	84.7	68.3	46.8	25.1	
		MCWB	18.2	28.5	40.1	48.9	55.8	62.0	66.5	65.1	58.8	49.3	35.7	21.7	
	2%	DB	15.0	28.3	49.4	66.4	80.1	88.5	90.6	86.8	78.4	61.1	40.8	19.6	
		MCWB	12.8	23.5	36.3	46.8	54.9	60.3	64.8	63.0	57.3	45.2	32.5	17.1	
	5%	DB	10.9	22.8	42.9	61.2	75.2	84.0	86.0	82.5	73.7	55.8	35.6	15.3	
		MCWB	9.1	19.3	32.7	43.9	52.8	59.5	63.9	61.7	54.9	42.2	29.1	13.3	
	10%	DB	7.0	18.3	37.5	56.4	70.6	79.7	81.9	78.3	68.8	51.0	30.9	11.8	
		MCWB	5.5	15.7	29.7	41.1	50.6	58.3	62.8	60.8	53.2	39.5	26.0	10.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.4	28.8	40.9	50.7	58.9	66.0	69.2	68.7	62.1	49.8	36.3	22.1
			MCDB	20.9	34.8	54.9	68.7	80.6	84.4	86.1	85.2	77.7	67.6	46.2	25.2
2%		WB	12.8	23.6	36.5	47.2	56.1	63.3	67.5	66.2	59.0	45.6	32.8	17.2	
		MCDB	15.0	28.0	48.3	65.4	76.2	80.7	83.5	78.7	71.6	59.1	40.6	19.6	
5%		WB	9.1	19.5	33.0	44.3	53.6	61.4	66.0	64.3	56.9	42.9	29.4	13.4	
		MCDB	10.8	22.8	42.3	60.2	72.4	78.5	80.7	75.9	68.9	54.7	35.2	15.2	
10%		WB	5.5	15.8	29.9	41.3	51.3	59.8	64.5	62.5	54.6	40.1	26.3	10.2	
		MCDB	7.0	18.2	37.2	55.7	68.8	75.9	77.7	74.2	66.9	50.2	30.6	11.8	
Mean Daily Temperature Range	5% DB	MDBR	17.8	22.9	24.2	25.1	27.5	24.6	21.9	23.1	25.4	23.5	20.5	16.8	
		MCDBR	21.0	27.1	29.9	33.4	35.4	32.0	28.6	30.2	31.8	30.8	25.6	18.8	
		MCWBR	19.2	22.9	19.6	18.5	16.4	12.3	10.1	11.8	15.4	17.9	18.3	17.0	
	5% WB	MCDBR	21.0	26.5	28.3	31.3	31.1	26.3	24.0	23.3	26.9	29.2	24.8	18.6	
		MCWBR	19.2	22.7	18.9	17.9	15.5	11.4	10.0	10.7	13.6	17.6	17.9	16.9	
Clear-Sky Solar Irradiance	taub	0.248	0.272	0.311	0.373	0.431	0.436	0.405	0.372	0.332	0.297	0.265	0.250		
	taud	2.323	2.286	2.268	2.156	2.027	2.131	2.289	2.355	2.404	2.427	2.385	2.304		
	Ebn at Noon	268	284	287	278	264	263	271	276	278	272	259	251		
	Edh at Noon	22	29	35	44	52	47	40	36	31	25	21	20		
All-Sky Solar Radiation	RadAvg	618	988	1460	1814	2063	2044	1949	1781	1449	1014	647	487		
	RadStd	38	58	57	75	95	137	92	99	93	58	33	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	-2.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page